

———— // ZPWR-MIDI-FX // ————

zpwr-midi-fx - Reference Manual

v0.30

MODULE REFERENCE

note-stream MIDI effect + generator modules

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MenkeTechnologies

———— MENKETECHNOLOGIES ————

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[0x00] PATCH MODEL

Built on the shared **zpwr-patch-core** (zpc) graph --- the same engine as zpwr-fx/zpwr-synth, instantiated on a note-event stream instead of audio samples. A dynamic set of blocks (add/remove/reorder), each with a module type, note-stream inputs + a scalar **Mod** input (unbounded summed cables per slot), up to six parameters, and one output. Inputs select any source; the two outputs (**Out A**, **Out B**) merge to the plugin's MIDI out. The graph is evaluated once per audio block in topological order, and note-offs are scheduled across block boundaries so nothing ever hangs.

An **EZ WIRE** button auto-wires MIDI In to your blocks' Out A so a beginner gets a working chain without touching cables; an **INIT** button unplugs every cable & mod while keeping the blocks, and **Blank** blanks the whole patch. The **INPUTS** column exposes every patchable source as a jack --- the note input, all scalar modulators (Random + performance/MPE controllers) and the soft keys --- so anything usable in the mod matrix can be cabled directly into a block.

Cables are drawn between jacks and can be dragged to re-patch. Right-click a cable for its editor: a **Level** control (scales note velocity on that connection --- **0** mutes it, and cable brightness/width follow the level), a **Colour** swatch, and **Disconnect**. Level is a live tweak; colour is cosmetic and persists with the patch.

Every control has a hover tooltip explaining what it does.

MIDI In [B1 Chord] [B2 Arp] Out A MIDI Out

[0x01] MODULE LIBRARY

66 **note-stream** modules --- harmony, sequencing, probability, routing, control sources, dynamics, tuning, MPE/voicing, plus a family of cellular-automaton sequencers (Game of Life, Brian's Brain, Langton's Ant). The table below is a representative selection; [docs/reference.html](https://docs.puredata.info/docs/reference.html) lists every block with its inputs and parameters, generated from the live registry so it never drifts.

Module	What it does
Chord	one key $\square$ a voiced chord (165 types, inversion, spread, octave-double, transpose, strum)
Arp	host-synced arpeggiator (9 play modes, divisions, gate, octaves, swing)
Scale	snap every note onto the nearest in-key pitch (20 scales)
Transpose	shift notes by semitones (modulatable)
Velocity	scale / offset note velocity (modulatable)
SeqEuclid	retrigger held notes on a Euclidean rhythm (Bjorklund)
Chance	per-note probability gate
Harmonize	stack up to three fixed intervals
Echo	MIDI delay with feedback (decaying repeats)
Merge	combine two note streams
LFO	scalar control source (sine / tri / saw / square)
Env	ADSR envelope follower, gated by held notes --- a scalar control source
Octave	add octave-up / octave-down copies
NoteFilter	pass only notes within a note + velocity range (key zone)
Mono	collapse to monophonic with note priority (last / lowest / highest)
Latch	toggle-hold notes --- sustain until re-pressed, ignore note-offs
Strum	spread simultaneous notes in time (up / down)
Quantize	snap note timing to a grid (rate + strength)
Channel	remap the output MIDI channel
FixedNote	force every note to one pitch (drum triggering)
Random	clock-driven random-note generator over a range
SampleHold	sample a scalar source on note triggers --- a control source
Slew	smooth a scalar source --- a control source
VelCurve	reshape velocity through a gamma curve
RandOctave	randomly shift notes by $\pm$ octaves (probability)
Humanize	jitter note timing and velocity for a played feel
SeqRatchet	split each note into N rapid retriggers over one division
KeySwitch	keyboard split --- one zone plays, the other holds back
Fold	octave-fold every note into a fixed [low, high] window
NoteLength	force every note to a fixed gate length
Accent	boost the velocity of every Nth note (downbeat)
VelClip	clamp note velocity into a [min, max] window
Invert	melodic inversion --- reflect notes around a pivot
Unison	layer each note as N copies, optional channel spread (MPE)
Ramp	velocity crescendo/decrescendo cycling over N notes

[0x02] MOD MATRIX

A dynamic list of routes, each mapping a **scalar source** to any **block parameter** with a depth. Sources:

- > **Soft Keys** (expandable pool of host-automatable macros; 16 active by default, +/- to add/remove, active count saved in plugin state)
- > **LFO** and **Env** block outputs
- > **Random**
- > **Performance controllers** --- Mod Wheel, Pitch Bend, Aftertouch, Velocity, Expression (CC11), Sustain (CC64)
- > **MPE** --- per-note Bend, Pressure and Slide (CC74), read from the most recently active note's channel

Routing (source / destination) rebuilds the graph; depth is a live, lock-free tweak. The same sources are available on each block's **Mod** input.

[0x03] PERFORM & GLOBAL

Stereo --- ☐ **STEREO** mirrors every block, cable and mod into an independent right-channel chain (**In R** ☐ **Out R**), kept in sync as you edit; knobs stay independent so you can dial width. ☐ **LOCK** additionally keeps the mirrored knobs locked to the left channel --- the lock is **bidirectional** (moving either the left knob or its clone moves the partner) and **offset-preserving** (the partner moves by the same delta, so a width you dialled in plain Stereo isn't reset to L=R). Locked clone blocks are dimmed.

PERFORM tab --- a macros-and-pads view with no patching, for live play:

- > **Preset Morph** --- a 4-corner XY pad (A/B/C/D) that bilinearly interpolates between four captured presets. Corner snapshots live on the processor; X/Y are reserved host params (**morphX/morphY**), so the morph is host-automatable. ☐ assigns a random preset to all four corners.
- > **Orb** --- angle picks one of 8 random ``scenes'' (per-macro offset vectors), distance from centre scales intensity. ☐ rolls new scenes; ☐ records orb motion while you drag; ☐ loops the recorded motion. Offsets are applied on top of the macro values captured when the gesture/playback started, and writes go through the soft-key params so they record as host automation.
- > **XY macro pads** --- each pad drives a pair of soft keys (X/Y). Per-pad **HOLD** keeps the dot where you drop it; **SPRING** snaps both axes back to centre on release. ☐ randomises a pad's two macros.
- > **Macro knobs** --- a dial per active soft key.
- > **Snapshots** --- 8 slots capturing the whole macro surface (per-plugin localStorage); click empty to save, filled to recall, right-click to clear.
- > **On-screen keyboard** --- global **Key + Scale** quantize of incoming notes and a **Chord** selector (Off / Oct / 5th / Maj / Min / Maj7 / Min7 / Sus4 / Power) that stacks extra intervals on each played key. The control band also exposes a global arp (mode / rate / latch) and a ☐ **RANDOMIZE** of all macro knobs. (midi-fx also has its own per-block **Arp** module with an independent latch --- distinct from this global arp.)

MIDI In response --- **PROGRAM** and **BANK** toggles (both **default ON**): incoming Program Change switches presets, with Bank Select (CC0 MSB / CC32 LSB) captured for the next Program Change. Both are saved in plugin state.

[0x04] ARCHITECTURE

```
libs/zpwr-patch-core/  the shared zpc graph (submodule) + the shared cyberpunk webui
dsp/                   JUCE-independent note engines (headless-testable)
                        MidiTypes · ChordDictionary · ChordEngine · Scale · Euclidean · Arpeggiator
                        MidiModules    - the 66 MIDI modules + note-stream signal policy, on zpc
src/                   JUCE plugin shell
                        PluginProcessor - APVTS soft keys, MIDI/MPE capture, transport-driven processBlock
                                      feeding zpc::PatchEngine<NoteStream>
                        PluginEditor   - thin host over zpc::WebEditor (shared catalog/preset bridge)
tests/ dsp_smoke.cpp (engines) · patchgraph_smoke.cpp (zpc graph, modules, JSON)
```

The note modules and the graph have no GUI dependency and are unit-tested without audio hardware or a plugin host. The patch (blocks, routing, mods) is zpc JSON the WebView edits and the plugin persists in its state.

[0x05] BUILD

JUCE 8.0.13 is vendored via CMake `FetchContent` (no system install required).

```
cmake -S . -B build
cmake --build build
```

Build a single format/target instead of all of them:

```
cmake --build build --target ZpwrMidIFx_Standalone # or _VST3 / _AU / _CLAP
```

On macOS the default build is a **universal binary** (`x86_64;arm64`) --- the VST3/AU/CLAP load on both Intel and Apple Silicon hosts. For a faster host-only dev build, configure a fresh build dir with `-DCMAKE_OSX_ARCHITECTURES=arm64` (the architecture is cached).

`scripts/build_pkg.sh` bundles the built universal VST3/AU/CLAP into a macOS `.pkg` that installs into the system plug-in folders (`/Library/Audio/Plug-Ins/{VST3,Components,CLAP}`) and the Standalone app into `/Applications`; run it after building the `ZpwrMidIFx_VST3` `ZpwrMidIFx_AU` `ZpwrMidIFx_CLAP` `ZpwrMidIFx_Standalone` targets `0 dist/ZpwrMidIFx-<version>.pkg`. Product name, version and bundle id are read from the build tree --- nothing is hardcoded.

Windows: builds `VST3` + `CLAP` (AU is macOS-only, auto-dropped); x64 and ARM64 are separate builds (no universal binary). `scripts/build_win.ps1` configures, builds each arch and stages to `dist/win<arch>\` --- `scripts/build_win.ps1 [-Arch x64|ARM64|both] [-Install]`. Requires Visual Studio 2022 (Desktop C++) + CMake; it installs the `Microsoft.Web.WebView2` NuGet package (needed by the WebView UI via `NEEDS_WEBVIEW2`) if missing. Validate with `pluginval` (VST3) and `clap-validator` (CLAP).

With `COPY_PLUGIN_AFTER_BUILD` the VST3/AU are installed to the user plugin folders automatically. To build against a local JUCE clone (offline):

```
cmake -S . -B build -DFETCHCONTENT_SOURCE_DIR_JUCE=/path/to/JUCE
```

Reference docs

`docs/reference.html` is the full module reference --- every note-stream block with its inputs and parameters, grouped by category. It is generated from the **live module registry** (the same catalog the plugin's UI shows), so it never drifts from the build. `docs/reference.pdf` is the same content paginated for print, built from that same HTML.

```
# regenerate docs/reference.html from the registry
cmake --build build --target gen_reference
build/gen_reference_artefacts/Debug/gen_reference docs/reference.html
# rebuild docs/reference.pdf (needs pandoc + xelatex) - also refreshes the HTML
scripts/reference_pdf.sh
```

The renderer (`zpc::renderReferenceHtml`) lives in `zpwr-patch-core` and is shared by all three plugins; per-block docs come from each block's `description/category` metadata.

[0x06] TEST

```
ctest --test-dir build --output-on-failure
```

Coverage: the chord dictionary and voicing engine, scale quantizer, Euclidean generator, arpeggiator traversal, and the full patch graph --- routing, pass-through, every module, the mod matrix (soft-key / aftertouch / envelope modulation) and JSON round-tripping.

Catalog at a glance

78 blocks shipped by **zpwr-midi-fx** across 9 categories, including 0 component-level analog-circuit models and 31 control / modulation (CV-badge) sources. This is the per-plugin subset of the shared zpwr-patch-core catalog.

Registry MIDI 78
Output jack Audio 0 MIDI 78 CV 0

By category (block count, largest first):

Sequencing.....	25	Control.....	5
Harmony.....	18	Voicing.....	4
Routing.....	9	Probability.....	1
Dynamics.....	8	Tuning.....	1
Modulation.....	7		

Reading this catalog (badge key)

An icon glyph precedes each block name; then its badges, then **in**: its input jacks and **out**: its output-jack type, then the description:

- > **[CAT]** category - **AMP** Amp, **ANA** Analog, **CTL** Control, **DLY** Delay, **DST** Distortion, **DYN** Dynamics, **ENV** Envelope, **FLT** Filter, **FX** Effect, **HRM** Harmony, **MOD** Modulation, **OSC** Oscillator, **PIT** Pitch, **PRB** Probability, **REV** Reverb, **RTE** Routing, **SEQ** Sequencing, **SPC** Spectral, **STR** Stereo, **TUN** Tuning, **UTL** Utility, **VOI** Voicing.
- > **[CV]** a control / modulation source (outputs CV, not audio).
- > **[CIRCUIT]** a component-level analog-circuit model (per-sample nodal / Newton solve).
- > **[FX]** **[SYNTH]** **[MIDI]** which **plugin(s)** ship the block - **zpwr-fx**, **zpwr-synth** (gets the whole audio pack) and/or **zpwr-midi-fx**.
- > **in**: the block's input jacks, and **out**: its output-jack signal type - **out: Audio**, **out: MIDI** (note stream) or **out: CV** (control / modulation).
- > **icon**: the glyph before each block name shows its kind at a glance:

	Oscillator		Filter		Delay
	Reverb		LFO / mod		Envelope
	Dynamics		Stereo		Pitch
	Spectral		Harmony		Sequencer
	Routing		Voicing		Effect
	Control		Probability		Tuning
	Utility		Guitar amp		Pedal / stompbox
	Speaker cabinet		Analog circuit		Other

MIDI (zpw-midi-fx) (78 blocks)

Control (5)

⌚ **ATProc** [CTL][CV][MIDI] in: Audio out: MIDI

Scales aftertouch, optionally converting it to a CC.

⌚ **BendProc** [CTL][CV][MIDI] in: Audio out: MIDI

Scales and/or inverts incoming pitch-bend.

⌚ **CC2Note** [CTL][CV][MIDI] in: Audio out: MIDI

A watched CC crossing a threshold triggers a note.

⌚ **CCMap** [CTL][CV][MIDI] in: Audio out: MIDI

Remaps a CC number and scales/offsets its value.

⌚ **Note2CC** [CTL][CV][MIDI] in: Audio out: MIDI

Emits a CC from each note (pitch or velocity).

Dynamics (8)

⌚ **Accent** [DYN][MIDI] in: Audio out: MIDI

Boosts velocity on every Nth note.

⌚ **Humanize** [DYN][MIDI] in: Audio out: MIDI

Adds random timing and velocity jitter for a human feel.

⌚ **Ramp** [DYN][MIDI] in: Audio out: MIDI

Velocity ramp over a run of notes, ascending or descending.

⌚ **VelClip** [DYN][MIDI] in: Audio out: MIDI

Clamps velocity between Min and Max.

⌚ **VelCompress** [DYN][MIDI] in: Audio out: MIDI

Velocity compressor: velocities above Threshold are pulled toward it by Ratio (1 = off, 8 = hard), narrowing the dynamic range smoothly instead of hard-clamping like VelClip.

T **VelCurve** [DYN] [MIDI] in: Audio out: MIDI

Reshapes velocity through a power curve.

T **VelInvert** [DYN] [MIDI] in: Audio out: MIDI

Flips the velocity scale around its midpoint (soft <-> loud) by Amount, so your accents become ghost notes and the ghosts become accents - a velocity mirror, blendable to taste.

T **Velocity** [DYN] [MIDI] in: Audio out: MIDI

Scales and offsets note velocity, with a modulation amount.

Harmony (18)

B **Bassline** [HRM] [MIDI] in: Audio out: MIDI

Follows the chord with a mono bass voice: the lowest held note dropped Octaves down, retriggered whenever the lowest note changes. Passes the original notes through and adds the bass beneath them.

B **Chord** [HRM] [MIDI] in: Audio out: MIDI

Turns each incoming note into a chord (165 types) with inversion, spread, octave doubling and strum.

B **ChordMem** [HRM] [MIDI] in: Audio out: MIDI

Each key plays a stored chord shape (semitone intervals from node config).

B **ChordProg** [HRM] [MIDI] in: Audio out: MIDI

Each note triggers the next chord in a stored root progression (node config).

B **ChordVoicing** [HRM] [MIDI] in: Audio out: MIDI

Re-voices each note into a jazz shape (open / shell / quartal / spread).

B **Counterpoint** [HRM] [MIDI] in: Audio out: MIDI

Adds a second voice in contrary motion - when the melody rises the counter voice falls and vice-versa - quantised to Root/Scale. Interval sets the starting separation.

B **Diatonic** [HRM] [MIDI] in: Audio out: MIDI

Transposes by scale DEGREES within Root/Scale (a third, a fifth...) instead of by raw semitones like Transpose - the shift always lands back in key. Mod Amt adds a modulatable degree offset.

⦿ **DiatonicChord** [HRM] [MIDI] in: Audio out: MIDI

Auto-harmoniser: builds the in-key chord on every note by stacking diatonic thirds (triad, 7th...) along Root/Scale, so single-finger melodies become correct chords in the key.

⦿ **Drone** [HRM] [MIDI] in: Audio out: MIDI

Doubles every note with a fixed pedal root.

⦿ **Glissando** [HRM] [MIDI] in: Audio out: MIDI

Fills a stepwise run between consecutive notes.

⦿ **Harmonize** [HRM] [MIDI] in: Audio out: MIDI

Adds up to three fixed-interval harmony voices to each note.

⦿ **Invert** [HRM] [MIDI] in: Audio out: MIDI

Mirrors pitch around a pivot note.

⦿ **MidiCluster** [HRM] [MIDI] in: Audio out: MIDI

Adds symmetric tone-cluster voices above and below each note.

⦿ **MidiFold** [HRM] [MIDI] in: Audio out: MIDI

Folds out-of-range notes back inside a Low..High window.

⦿ **NeoRiemann** [HRM] [MIDI] in: Audio out: MIDI

Neo-Riemannian triad transforms: builds a major/minor triad on each note and applies P (parallel), L (leading-tone) or R (relative) - the maximally-smooth chord moves of late-Romantic and film harmony, each sharing two common tones with the input.

⦿ **Octave** [HRM] [MIDI] in: Audio out: MIDI

Doubles each note up and/or down by whole octaves.

⦿ **Scale** [HRM] [MIDI] in: Audio out: MIDI

Quantizes incoming notes to the nearest degree of the chosen root and scale.

⦿ **Transpose** [HRM] [MIDI] in: Audio out: MIDI

Shifts every note by a fixed semitone amount plus a modulatable offset.

Modulation (7)

🌀 **Generative** [MOD][MIDI] in: Audio out: MIDI

Probabilistic melodic random walk synced to a rate.

🌀 **MidiEnv** [MOD][MIDI] in: Audio out: MIDI

ADSR envelope triggered by the note gate, on the scalar control output.

🌀 **MidiLFO** [MOD][MIDI] in: Audio out: MIDI

Low-frequency modulation source (sine/tri/saw/square) on the scalar control output.

🌀 **MidiRandom** [MOD][MIDI] in: Audio out: MIDI

Generates random notes within a pitch range at a synced rate.

🌀 **MidiSampleHold** [MOD][MIDI] in: Audio out: MIDI

Samples and holds the incoming control value on each note.

🌀 **MidiSlew** [MOD][MIDI] in: Audio out: MIDI

Smooths (glides) the control value over a set time.

🌀 **RandOctave** [MOD][MIDI] in: Audio out: MIDI

Randomly shifts notes by up to +/- Range octaves with a chance.

Probability (1)

🎲 **Chance** [PRB][CV][MIDI] in: Audio out: MIDI

Probabilistic gate: lets each note through with the set probability.

Routing (9)

➤ **Channel** [RTE][MIDI] in: Audio out: MIDI

Reassigns notes to a fixed MIDI channel.

➤ **FixedNote** [RTE][MIDI] in: Audio out: MIDI

Forces every note to a single fixed pitch (drum / trigger).

➤ **KeySwitch** [RTE][MIDI] in: Audio out: MIDI

Splits the keyboard at a note; one zone passes, the other is filtered.

➤ **Merge** [RTE] [MIDI] in: Audio out: MIDI

Passes both note-stream inputs through, merged into one stream.

➤ **MidiLatch** [RTE] [MIDI] in: Audio out: MIDI

Holds notes after release until the next note arrives.

➤ **NoteFilter** [RTE] [MIDI] in: Audio out: MIDI

Passes only notes inside a key and velocity window.

➤ **NoteLimit** [RTE] [MIDI] in: Audio out: MIDI

Polyphony cap: never lets more than Max notes sound at once, stealing the oldest (FIFO note-off) when a new note would exceed the limit - tames dense chords and runaway generators.

➤ **Split** [RTE] [MIDI] in: Audio out: MIDI

Keyboard split: notes below the split point transpose by Low, the rest by High.

➤ **Transformer** [RTE] [MIDI] in: Audio out: MIDI

Rule: notes within a range are transposed and velocity-scaled.

Sequencing (25)

■ ■ ■ **Appoggiatura** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Leaning grace note: each struck note first sounds a long auxiliary (Interval away) that takes Time from the principal, which then sustains - the expressive long grace, unlike the quick Flam or Mordent.

■ ■ ■ **Arp** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Arpeggiator: cycles held notes through a pattern at a synced rate with gate, octaves, swing and step count. Latch holds the chord so it keeps arpeggiating after the keys are released (the next key starts a new chord).

■ ■ ■ **Cascade** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Each struck note fans out into a timed run of Steps notes climbing (or falling) the scale from it - an arpeggiated flourish fired per key, unlike Arp which cycles the whole held chord.

■ ■ ■ **Echo** [SEQ] [CV] [MIDI] in: Audio out: MIDI

MIDI delay: repeats notes at a synced time with velocity feedback decay.

■□□ **Flam** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Precedes every note with a quiet grace note, delaying the main hit so the grace leads it - the classic drum flam.

■□□ **MidiBrianBrain** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Brian's Brain MIDI sequencer: an 8x8 three-state grid (ready / firing / dying) evolves under Brian's Brain rules - a ready cell ignites only with exactly two firing neighbours, firing cells pass to dying, dying cells reset. Each clock step plays a note for every firing cell in the next column (row maps to a scale degree from Root/Scale); a full 8-step sweep advances one generation. The refractory state keeps it restless and sparkly where Game of Life settles. Density sets the random fill. The MIDI-note counterpart to the BrianBrain CV block.

■□□ **MidiGameOfLife** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Conway's Game of Life MIDI sequencer: an 8x8 cell grid evolves under the B3/S23 rule; each clock step reads the next column and plays a note for every live cell (row maps to a scale degree from Root/Scale), and a full 8-step sweep advances one generation. Gliders, blinkers and still-lives become ever-shifting melodic and harmonic patterns. Density sets the random fill when the grid (re)seeds. The MIDI-note counterpart to the GameOfLife CV block.

■□□ **Midilangton** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Langton's Ant MIDI melody: a single turmite walks an 8x8 grid - turning right on a white cell, left on a black one, flipping each cell as it leaves, then stepping forward. Each clock advances the ant Steps cells and plays one note for its current row (mapped to a scale degree from Root/Scale), so the melody scribbles chaotically then locks into the famous periodic 'highway'. Emergent order from one trivial rule - a single moving agent, not a population.

■□□ **MidiQuantize** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Snaps note starts to a synced grid by a strength amount.

■□□ **MidiTranceGate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Chops the held notes with a 16-step on/off Pattern at a synced Rate: every 'on' step retriggers all held notes, gated to a fraction of the step. The rhythmic gate behind the trance sound - pattern-driven, where Roll is a fixed pulse.

■□□ **MidiTuring** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Turing-machine sequencer (Music Thing style): a circular shift register clocks once per step and the held notes fire when the read bit is 1. Change is the probability the recycled bit flips - 0 locks an evolving loop, 1 fully randomises, between = slowly mutating melodies.

■□□ **Mordent** [SEQ] [CV] [MIDI] in: Audio out: MIDI

One-shot ornament: each struck note plays principal -> auxiliary -> principal at the attack, then sustains, the single-flick counterpart to the continuous Trill.

■□□ **NoteLength** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Forces a fixed note duration (gate length).

■□□ **NoteOffDelay** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Extends each note's gate by delaying its note-off.

■□□ **PatternGate** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Gates note-ons through a 16-step on/off pattern (bitmask).

■□□ **Polymeter** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Two step lanes of different lengths (LenA, LenB) clock together over the held notes, phasing against each other so the combined pattern only repeats after lcm(LenA, LenB) steps - instant polymetric movement from one chord.

■□□ **Ricochet** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Accelerating bouncing retriggers with decaying velocity.

■□□ **Roll** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Drum roll / buzz: retriggers every held note at Rate for as long as it is held, each hit gated to a fraction of the interval.

■□□ **SeqEuclid** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Euclidean rhythm gate: spreads Pulses evenly across Steps (rotatable) at a synced rate.

■□□ **SeqRatchet** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Retriggers each note Count times within a synced step.

■□□ **StepSeqMidi** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Melodic step sequencer - semitone offsets (node config) clock the held root.

■□□ **Strum** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Spreads a chord's notes over time, up or down, like a guitar strum.

■□□ **Swing** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Pushes off-beat eighth-notes later for a swing/shuffle feel.

■□□ **Trill** [SEQ] [CV] [MIDI] in: Audio out: MIDI

Keyboard trill: while a note is held, alternates rapidly between it and the note Interval semitones above, at Rate.

■□□ **Turn**[seq][cv][MIDI] in: Audio out: MIDI

Gruppetto turn: a four-step ornament at the attack - upper auxiliary, principal, lower auxiliary, then the principal held. Time sets each step, Interval the neighbour distance.

Tuning (1)

U **Microtuner**[TUN][MIDI] in: Audio out: MIDI

Stretched-octave microtuning via per-note pitch detune.

Voicing (4)

⊗ **MidiUnison**[VOI][MIDI] in: Audio out: MIDI

Doubles each note into multiple voices, optionally spread across channels.

⊗ **Mono**[VOI][MIDI] in: Audio out: MIDI

Monophonic note priority (last / lowest / highest).

⊗ **MPEConvert**[VOI][MIDI] in: Audio out: MIDI

Assigns each note its own rotating channel for per-note MPE expression.

⊗ **Sustain**[VOI][MIDI] in: Audio out: MIDI

Sustain-pedal simulation: while Hold is engaged, note-offs are captured so the notes keep sounding; when Hold drops, every held note releases at once. Hold is a parameter, so a CC, an LFO or automation can play the pedal.

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